Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/19/2024
PI Name:	Yun, Sanghee Ph.D.		
Project Title:	Exposure of Male and Female Mice to 33-Particle Galactic Cosmic Radiation (GCR): Negative Impact on Translationally-relevant Hippocampal-based Cognition and Prevention of GCR-induced Deficits by Acute Treatment with the Anti-inflammatory CDDO-EA		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4318	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	10/01/2021	**End Date:**	09/30/2023
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per S. Mack/JSC and NSSC information (Ed., 5/22/23) NOTE: End date changed to 03/31/2023 per NSSC information (Ed., 10/12/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Eisch, Amelia Ph.D. (Children's Hospital of Philadelphia)		
Grant/Contract No.:	80NSSC21K0814		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: Galactic Cosmic Radiation (GCR) is an unavoidable factor in deep space that may affect astronaut health and mission success. To date, ground-based studies on the cognitive impact of GCR have exposed rodents to a single high-energy particle or a combination of a few particles. It is unknown how a complex, multiple-ion GCR spectrum -- as will be encountered during a Mars mission -- influences rodent cognition. It is also important to identify countermeasures that can reduce risk to crew health or performance. Preclinical and clinical studies suggest the anti-inflammatory compound CDDO-EA mediates its therapeutic effects in combating lung disease by upregulating the key antioxidant transcription factor Nrf2. Notably, in mouse models of neurological disorders, CDDO-EA ameliorates striatal- and hippocampal-dependent cognitive and cellular deficits. Thus, it is timely and urgent to assess if a complex GCR spectrum decreases rodent cognition and if CDDO-EA mitigates such changes. Hypothesis: Exposure to a complex GCR mixture will decrease translationally-relevant cognitive function and negatively impact associated cellular and molecular processes in female and male mice, and CDDO-EA will mitigate these changes. Aim 1. We will determine the effects of (1a) 33-particle GCR (33-GCR) on hippocampal dentate gyrus (DG) function (pattern separation) and prefrontal cortex (PFC) functions (cognitive flexibility, extinction of stimulus-response) in 6-month-old mature male C57BL/6J mice and (1b) CDDO-EA on 33-GCR-induced cognitive changes. Aim 2. We will also define DG neurogenesis and DG and PFC inflammation in female and male mice exposed to 33-GCR or Sham with and without CDDO-EA. Methods: Mature C57BL/6J male and female mice will receive CDDO-EA (400 mg/kg, n=80) or Control Chow (Control; n=80) for 5 days (3 days before, during, and 1 day after 33-GCR). Mice will receive either Sham or whole-body 33-GCR (~73% 1H, 18% 4He, 2.4% 12C, 3.1% 16O, 1.6% 28Si, 0.9% 48Ti, 0.8% 56Fe; 750mGy) with the 4 groups being Control/Sham, Control/33-GCR, CDDO-EA/Sham, CDDO-EA/33-GCR (n=16/group). To complement pilot data with female mice, in Aim 1, 2-mon post-irradiation male mice will undergo general touchscreen training, location discrimination (LD), and acquisition and extinction of stimulus-response. In Aim 2, brains will be collected from male and female mice at three timepoints post-33-GCR: 3 days, 2-mon, and 11-mon. 2- and 11-mon groups will be given bromodeoxyuridine (BrdU, i.p. 180 mg/kg) 2 hrs prior to perfusion and indices of neurogenesis and inflammation will be quantified via unbiased stereology. In the 3-day group, protein expression of Nrf2/ARE signaling targets will be quantified in the DG and PFC. Aim 1 deliverables include measures of LD (session length, # trials, % correct, etc.) and acquisition and extinction (session length, # responses, # omissions, days to completion) in male mice, and the impact of 33-GCR vs. Sham and CDDO-EA vs. Control Chow on these measures. Aim 2 deliverables include indices of DG neurogenesis (# of Ki67-, DCX-, BrdU-immunoreactive [+] cells) and DG and PFC inflammation (# of Iba-1+, Glial fibrillary acidic protein (GFAP) + cells)), and protein expression of signaling mediators downstream from nuclear factor erythroid 2–related factor 2 (Nrf2)/antioxidant response element (ARE). Significance: It is imperative to understand how a mission-relevant GCR mixture influences cognition, and determination of GCR-induced cellular and molecular changes enables identification of appropriate countermeasures. The proposed work in male and female mice will complement our already-completed cognitive study in female mice (manuscript in preparation). Given the similarity between cognitive function in mice and humans, the resulting data will allow more accurate estimation of space-radiation risks to both male and female astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is imperative to understand how a mission-relevant GCR mixture influences cognition, and determination of GCR-induced cellular and molecular changes enables identification of appropriate countermeasures. Our work in male and female mice will complement our already-completed cognitive study in female mice (manuscript in preparation). Given the similarity between cognitive function in mice and humans, the resulting data will allow more accurate estimation of space-radiation risks to both male and female astronauts. Our study will elucidate what the acute and late CNS risk are at the molecular, cellular, and behavioral levels for astronauts exposed to GCR.
	Dr. Sanghee Yun, a Principal Investigator, provides the day-to-day supervision of the team. Dr. Amelia Eisch, as Co-investigator, and postdoctoral fellows Dr. Fred Kiffer and Dr. Lori Colon, help and advise Dr. Yun based on their previous NASA runs; they completed NSRL 22A together with Dr. Yun at Brookhaven National Lab (BNL). At Children's Hospital of Philadelphia (CHOP), the current and past research staff that spearhead the behavioral tests (spontaneous location recognition/SLR and touchscreen tests) and cellular experiments include Dr. Kiffer and University of Pennsylvania (PENN) work-study students Raymon Shi, Priya Kumar, Harley Haas, Grace Bancroft, and Amishi Mahajan and summer intern program students, Stephanie Uroda, Hannah An, Elise Wallen-Friedman (PENN), and Chris DeSantis (Yale). Experiments were completed at CHOP on female and male mice from NSRL 22A (supported by Eisch's Institutional Animal Care and Use Committee/IACUC protocols from BNL and CHOP). In addition, we completed the NASA Augmentation award project (in vivo recording of glutamatergic cell activity in the dentate gyrus during SLR test using fiber photometry months after exposure to 33-GCR). Our lab has made progress on our NASA-funded research as described below: NASA Human Exploration Research Opportunities (HERO) (PI: Yun)-funded publications (7): 1. Tran FH, Spears SL, Ahn KJ, Eisch AJ, Yun S. Does chronic systemic injection of the DREADD agonists clozapine-N-oxide or Compound 21 change behavior relevant to locomotion, exploration, anxiety, and depression in male non-DREADD-expressing mice? Neurosci Lett. 2020 Nov 20;739:135432. 2. Clark LR, Yun S, Acquah NK, Kumar PL, Metheny HE, Paixao RC, Cohen AS, Eisch AJ. Mild traumatic brain injury induces transient, sequential increases in proliferation, neuroblasts/immature neurons, and cell survival: A TimeCourse study in the male mouse dentate gyrus. Front Neurosci. 2021 Jan 7;14:612749. 3. Soler I, Yun S, Reynolds RP, Whoolery CW, Tran FH, Kumar PL, Rong Y, DeSalle MJ, Gibson AD, Stowe AM, Kiffer FC. Multi-domain touchscreen-based cognitive assessment of C57BL/6J female mice shows whole-body exposure to 56Fe particle space radiation in maturity improves discrimination learning yet impairs stimulus-response rule-based habit learning. Front Behav Neurosci. 2021 Oct 11;15:722780.

Task Progress:	4. Kiffer FC, Luitel K, Tran FH, Patel RA, Guzman CS, Soler I, Xiao R, Shay JW, Yun S, Eisch AJ. Effects of a 33-ion sequential beam galactic cosmic ray analog on male mouse behavior and evaluation of CDDO-EA as a radiation countermeasure. Behav Brain Res. 2022 Feb 15;419:113677. 5. Cotter KM, Bancroft GL, Haas HA, Shi R, Clarkson AN, Croxall ME, Stowe AM, Yun S, Eisch AJ. Use of an automated mouse touchscreen platform for quantification of cognitive deficits after central nervous system injury. In: Karamyan, VT, Stowe, AM, eds. Neural Repair. Methods Mol Biol. Vol. 2616. New York, NY: Humana, 2023. p. 279-326. 6. Guez-Barber D, Colon LM, Raphael D, Wragan MA, Yun S, Eisch AJ. Female and male microglia are not different in the dentate gyrus of postnatal day 10 mice. Neurosci Lett. 2023 Apr 23;803:137171. 7. Yun S, Soler I, Tran FH, Haas HA, Shi R, Bancroft GL, Suarez M, de Santis CR, Reynolds RP, Eisch AJ. Behavioral pattern separation and cognitive flexibility are enhanced in a mouse model of increased lateral entorhinal cortex-dentate gyrus circuit activity. Front Behav Neurosci. 2023 Jun 1;17:1151877. NASA HERO (PI Yun)-funded publications in prep (3): 1. Yun S, Kiffer FC, Guzman CS, Soler I, Bancroft GL, Haas HA, Shi R, Patel R, Lara-Jiménez J, Kumar PL, Tran FH, Ahn KJ, Rong Y, Luitel K, Shay JW, Eisch AJ. Contemporaneous ingestion of the antioxidant CDDO-EA prevents galactic cosmic ray-induced decline in simple stimulus-response learning and hippocampal neurogenesis in female C57BL/6J mice. 2. Kiffer FC, Hamandi A, Russo A, Johnson N, Roura GC, Gautreaux E, Bathwal M, Yang E, Gawon K, Kogawa L, Stanislaus D, Sil A, Chen A, Shi R, Gillis S, Barber DG, Yun S, Eisch AJ. Effects of a fractionated whole-body exposure to 56Fe high-energy particles on social behavior and neural activation in female and male C57BL/6J mice. 3. Yun S*, Mahajan A, Bancroft GL, Haas HA, Wallen-Friedman E, de Santis CR, Vasisht S, Takano H, Kiffer, FC, Eisch AJ. Exposure of Male and Female Mice to 33-Particle Galactic Cosmic Radiation (GCR): Impact on behavioral pattern separation and Pavlovian learning, and Prevention of GCR-induced Deficits by Acute Treatment with the Anti-inflammatory CDDO-EA *Corresponding author. NASA HERO (PI: Yun) project-related findings: We examined the impact of exposure to 33-particle galactic cosmic radiation (GCR) on cognitive performance in fully mature male and female mice and mitigation of 2-cyano-3, 12-dioxooleana-1, 9-dien-28-oic acid (CDDO)-ethylamide (CDDO-EA), an antioxidant provided to the mice around the time of irradiation. We purchased male (n=124) and female (n=48) wildtype (C57BL/6J) mice for NSRL 22A (April 20, 2022). The variables for this experiment were irradiation (Sham and 33-GCR) and antioxidant (Veh or CDDO-EA), resulting in four groups of mice/sex (Sham/Veh, 33-GCR/Veh, Sham/CDDO-EA, 33-GCR/CDDO-EA; n=12/group/sex). After BNL, the mice were returned to CHOP. Female and male mice then were assessed for: monthly weight checks, home cage locomotion activity monitoring (18hrs), spontaneous location recognition test (a pattern separation test), training and testing on an automated touchscreen platform (Pavlovian learning) and elevated plus maze (a test for anxiety). In short, we found the acute exposure to 33-GCR (0.75 Gy, 1hr 40min) at 6 mon-old increases behavioral pattern separation– a type of high-resolution memory for discriminating similar stimuli– in male mice but not female mice. Also, administration of an antioxidant, CDDO-EA (4 mg/kg i.p for 3 days: the day before, on the day, and a day after IRR) normalizes male behavioral pattern separation. This test is highly sensitive to the dentate gyrus–a hippocampal subregion. Dentate gyrus (DG) activity was measured during the behavioral pattern separation via an indirect measurement of DG granular cell activity. We found that exposure to the 33-GCR decreases DG cell activity during exploration in similar stimuli which implies neural mechanisms of 33-GCR-induced behavioral pattern separation changes. Also, we tested association learning, a type of Pavlovian learning. We found that 33-GCR alone transiently decreases “sign tracking” behavior–approaches to the reward-related condition stimuli during Acq. in males, not females. CDDO-EA itself (in Sham, non-IRR mice) does not change “sign tracking” during Acq. in both males and females. Unexpectedly, 33-GCR exposure increased reversal learning during reverse learning to the acquired learning in male mice. However, female approaches to CS+ are unchanged during reversal learning. There are several implications of our study. 1) The exposure to a mission-relevant dose of 33-GCR may increase pattern separation ability in male astronauts, but not female astronauts after a mission. 2) CDDO-EA ameliorates cognitive deficits after a mission. These rodent findings warrant additional study with radiation parameters that decrease cognition. 3) Exposure to a mission-relevant dose of GCR has a sex-specific influence on different forms of discrimination and associative learning. Thus, more work with 33-GCR is warranted. For example, as our study of real-time measurement of the DG activity in freely behaving animals gives insight into cellular mechanisms underlying 33-GCR-induced improvement of behavioral pattern separation, fMRI study in astronauts during or after their mission will be warranted. So far, we have shared these data each year at the NASA Human Research Program Investigators' Workshop (HRP IWS) meeting and Radiation Research in order to receive feedback on our approach and interpretations. The resulting data are the focus of several manuscripts we have in preparation/revision, and Dr. Yun (PI of this HERO grant) is first and/or senior author on each manuscript.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Guez-Barber D, Colon LM, Raphael D, Wragan MA, Yun S, Eisch AJ. "Female and male microglia are not different in the dentate gyrus of postnatal day 10 mice." Neurosci Lett. 2023 Apr 23;803:137171. https://doi.org/10.1016/j.neulet.2023.137171 ; PMID: 36898652 , Apr-2023
Articles in Peer-reviewed Journals	Yun S, Soler I, Tran FH, Haas HA, Shi R, Bancroft GL, Suarez M, de Santis CR, Reynolds RP, Eisch AJ. "Behavioral pattern separation and cognitive flexibility are enhanced in a mouse model of increased lateral entorhinal cortex-dentate gyrus circuit activity. " Front Behav Neurosci. 2023 Jun 1;17:1151877.https://doi.org/10.3389/fnbeh.2023.1151877 ; PMID: 37324519; PMCID: PMC10267474 , Jun-2023

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Articles in Peer-reviewed Journals	Guez-Barber D, Eisch A, Cristancho AG. "Developmental brain injury and social determinants of health: Opportunities to combine preclinical models for mechanistic insights into recovery." Dev Neurosci. 2023 April 20;45(5):255-267. https://doi.org/10.1159/000530745 ; PMID: 37080174; PMCID: PMC10614252 , Apr-2023
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Awards	Haas H. "Human Roadmap program Augment Award/The National Aeronautics and Space Administration (NASA), February 2021. (PI: Sanghee Yun)." Feb-2021
Awards	Yun S. "2022 Radiation Research Society Conference Travel Award, October 2022." Oct-2022